THE UNIVERSITY OF AUCKLAND

SEMESTER TWO 2023
Campus: City

MECHANICAL ENGINEERING

Biomechatronic Systems

(Time Allowed: TWO hours)

NOTE: Answer ALL THREE questions.
For multiple choice questions, please circle the correct answer.
For true false questions, please circle the correct answer.

Surname: ……..…….. Forenames: ……

University of Auckland ID number: ……

Question 1 (30 marks)

MULTIPLE CHOICE QUESTIONS. QUESTIONS 1.1 – 1.10

1.1. Which one of the following statements is correct?

a. A healthy adult human typically undergoes 9 sleep cycles overnight.
b. A healthy adult human typically undergoes 7 sleep cycles overnight.
c. A healthy adult human typically undergoes 3 sleep cycles overnight.
d. A healthy adult human typically undergoes 5 sleep cycles overnight.

(2 marks)

1.2. Which one of the following statements is correct?

a. In humans, sleep episodes are circadian and sleep cycles are ultradian.
b. In humans, sleep episodes are ultradian and sleep cycles are circadian.
c. At sleep onset, all animals typically transition from wakefulness to rapid eye movement (REM) sleep.
d. At sleep onset, humans typically transition from wakefulness to rapid eye movement (REM) sleep.

(2 marks)

1.3. Which one of the following statements regarding rapid eye movement (REM) and non-rapid eye movement (NREM) sleep is correct?

a. In REM sleep, electromyogram (EMG) activity is high.
b. Eye movements occur only during REM sleep but typically not during NREM sleep.
c. Sleep spindles are characteristic of REM sleep.
d. REM sleep is associated with fast, desynchronised electroencephalogram (EEG) activity.

(2 marks)

1.4. Which one of the following lists shows common sleep disorders in order of increasing prevalence in the general population?

a. insomnia, narcolepsy, restless legs syndrome, obstructive sleep apnoea
b. narcolepsy, restless legs syndrome, obstructive sleep apnoea, insomnia
c. obstructive sleep apnoea, insomnia, narcolepsy, restless legs syndrome
d. narcolepsy, obstructive sleep apnoea, insomnia, restless legs syndrome

(2 marks)

1.5. Which one of the following statements is true?

a. An apnoea-hypopnea index (AHI) of 25 events per hour is indicative of severe sleep apnoea.
b. Old age is a risk factor for central sleep apnoea.
c. Elongation of the central soft palate is a risk factor in central sleep apnoea.
d. Compared to males, females are at greater risk of obstructive sleep apnoea.

(2 marks)

1.6. Which one of the following statements about continuous positive airway pressure (CPAP) is correct?

a. CPAP may produce slow-wave sleep rebound or REM sleep rebound.
b. CPAP usually produces increased sleep fragmentation.
c. CPAP is the only effective treatment available for obstructive sleep apnoea.
d. CPAP is a treatment for obstructive and central sleep apnoea.

(2 marks)

1.7. Which one of the following statements provides the best definition of sleep?

a. Sleep is a state during which behaviour is absent.
b. Sleep is a reversible state of perceptual disengagement and unresponsiveness.
c. Sleep is a state during which the muscles of the eyeballs are usually in motion behind the eyelids. People emerging from this state will generally report that they were dreaming.
d. Sleep is a behavioural state of mostly unrousable unconsciousness.

(2 marks)

1.8. Which one of the following statements is NOT correct?

a. In sleep, episodic bodily movements are associated with rapid eye movement (REM) sleep.
b. Sleep impairs memory of the five minutes preceding sleep onset.
c. A healthy adult human typically spends more time in sleep stage N2 than rapid eye movement (REM) sleep.
d. In healthy adult humans, rapid eye movement (REM) sleep accounts for approximately 40% of sleep.

(2 marks)

1.9. Which one of the following statements about the Electroencephalogram (EEG) is correct?

a. EEG activity at frequencies greater than 50 Hz contains little information about brain activity.
b. EEG activity at 3 Hz with peak amplitude near 20 microvolts would be described by a clinician as "fast, desynchronised".
c. Alpha-band activity is comprised of lower frequencies than delta-band activity.
d. In overnight sleep studies, it is typical to use 32 EEG electrodes.

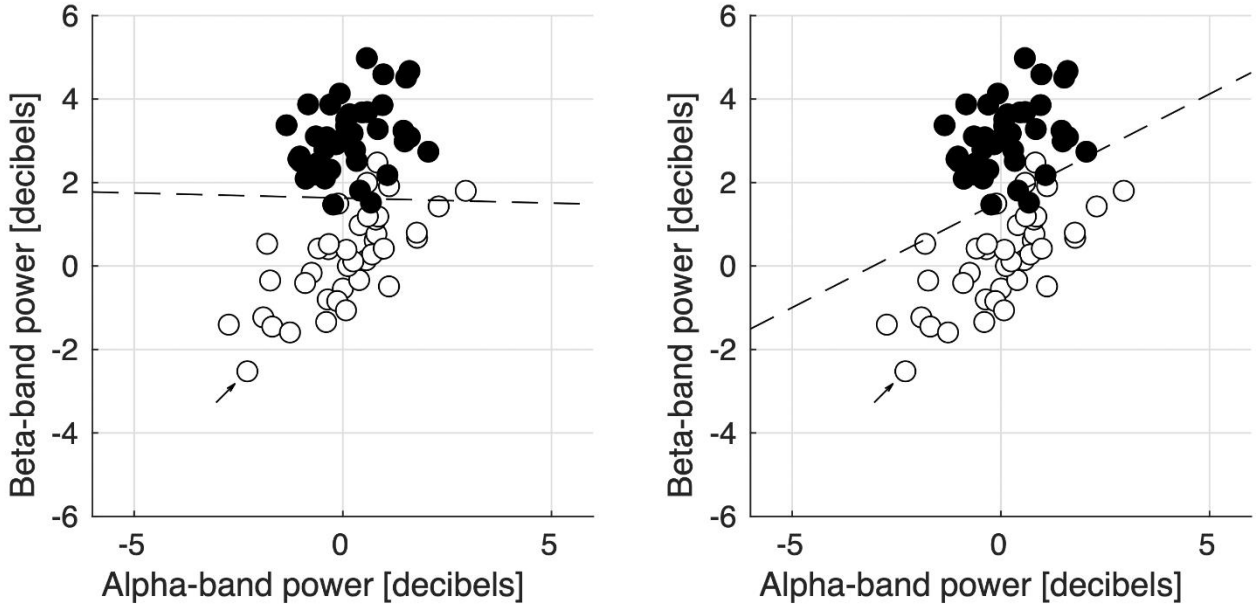
(2 marks)

1.10. Slow eye movements and a marked decreased in electroencephalogram (EEG) alpha-band activity are characteristic of which one of the following sleep stage transitions?

a. Transition from rapid eye movement (REM) sleep to non-REM sleep.
b. Transition from slow-wave sleep (SWS) to rapid eye movement (REM) sleep.
c. Transition from wakefulness to sleep stage 1.
d. Transition from sleep stage N2 to N3.

(2 marks)

Questions 1.11 to 1.15 pertain to the following plots:



In the leftmost plot, Student A has represented their electroencephalogram (EEG) recordings in a feature space, and has fit a classification model (henceforth, "model 1") to discriminate two samples: "sample 1" (open circles) and "sample 2" (closed circles). The dashed line shows the decision boundary. (The observation indicated by the arrow is discussed in a question below.)

In the rightmost plot, Student B has fit a classification model ("model 2") to discriminate the same two samples. Graphical conventions are the same here as in the leftmost plot.

Both model 1 and 2 are Fisher's linear discriminant analysis (LDA) models.

1.11. Are the samples linearly separable?

(2 marks)

1.12. If Student A removed the observation indicated by the arrow, would their fit change? Explain your answer.

(2 marks)

1.13. During model fitting, Student A and B made different assumptions about the covariance of alpha- and beta-band power. What were the assumptions? Explain your answer.

(2 marks)

1.14. Write pseudocode, or concise plain English, which illustrates how to use cross-validation to compare the predictive performance of model 1 and 2.

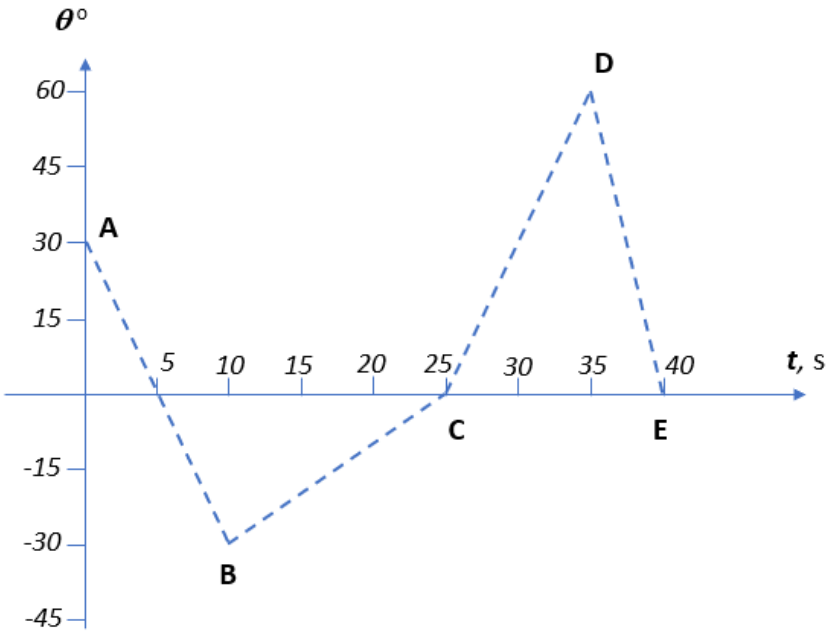
(2 marks)

1.15. Student A and B collaborate to collect more data. When used to classify these new data, would you expect model 1 to outperform model 2, or vice versa? Explain your answer.

(2 marks)

Question 2 (40 marks)

2.1 A robotic joint is at rest at position **A,** before rotating via positions **B**, **C** and **D** until coming to rest at position **E**. The angle and time of each position is shown in the figure below.



i) Find appropriate values for the angular velocities at via points **B**, **C** and **D**.

(3 marks)

ii) Find the angular displacement $\theta(t)$ for segment **BC** using a polynomial, based on i).

(6 marks)

iii) Sketch $\theta(t)$, $\dot{\theta}(t)$ and $\ddot{\theta}(t)$ plots for the motion of segment **BC** found in ii).

(5 marks)

Formulae below might be needed to answer Question 2.1:

$\theta(t) = a_0 + a_1 t + a_2 t^2 + a_3 t^3$

$\dot{\theta}(t) = a_1 + 2a_2 t + 3a_3 t^2$

$\ddot{\theta}(t) = 2a_2 + 6a_3 t$

$a_0 = \theta_0$

$a_1 = \dot{\theta}_0$

$a_2 = \frac{3}{t_f^2}(\theta_f - \theta_0) - \frac{2}{t_f}\dot{\theta}_0 - \frac{1}{t_f}\dot{\theta}_f$

$a_3 = -\frac{2}{t_f^3}(\theta_f - \theta_0) + \frac{1}{t_f^2}(\dot{\theta}_f + \dot{\theta}_0)$

$\theta_0 = a_0$

$\theta_f = a_0 + a_1 t_f + a_2 t_f^2 + a_3 t_f^3$

$\dot{\theta}_0 = a_1$

$\dot{\theta}_f = a_1 + 2a_2 t_f + 3a_3 t_f^2$

2.2 The following questions are in the context of genetic algorithms.

i) What is an appropriate value for p_c, a pre-set crossover probability?

(2 marks)

ii) What is an appropriate value for p_m, a pre-set mutation probability?

(2 marks)

iii) Two parental chromosomes are given below. Perform a single-point crossover with crossover site at 2 and give the two resultant child chromosomes.

0 1 1 0 1 0 1 1 0
1 1 1 1 0 0 1 0 1

(2 marks)

iv) Name two major differences between a genetic algorithm and other traditional search algorithms.

(2 marks)

v) Briefly describe how a mutation operation is performed.

(2 marks)

2.3. The following questions are in the context of biomechanics.

i) State what TMJ is an abbreviation for and the key motion features of a TMJ. (2 marks)

ii) What are the three motion phases in the chewing cycle of the mandible? (2 marks)

iii) What is the Posselt envelope? (2 marks)

iv) What is a sarcomere and in what unit is its length described?

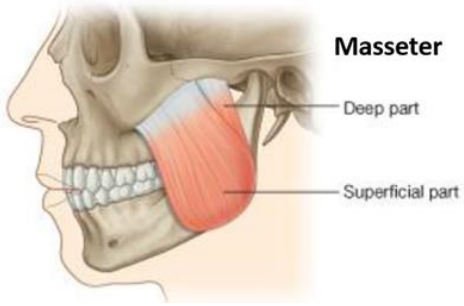
(2 marks)

v) Describe how a human musculoskeletal system is voluntarily controlled according to the servo hypothesis, with the aid of a schematic diagram.

(4 marks)

vi) Describe with the aid of a sketch the biomechanical model of the superficial and deep parts of the masseter muscle, as shown below.

(4 marks)



Question 3 (30 marks)

3.1 In the context of underactuated, compliant, humanlike robot hands with passive extension, please answer the following. Each question is worth 1 mark.

i) The range of the post-contact reconfiguration of the hand-object system depends on the lengths of the finger links.

(1 mark)

- True
- False

ii) The free space reconfiguration of a hand during actuation depends on the stiffness of the finger joints.

(1 mark)

- True
- False

iii) The free space reconfiguration of a hand depends on the weight of the different phalanges of the finger.

(1 mark)

- True
- False

iv) The post-contact reconfiguration of the hand-object system depends on hand orientation.

(1 mark)

- True
- False

v) The free space reconfiguration of the hand depends on the orientation of the hand but NOT on the position if the orientation is fixed.

(1 mark)

- True
- False

vi) The post-contact reconfiguration of the hand-object system is NOT affected by the grasped object material.

(1 mark)

- True
- False

vii) The post-contact reconfiguration of the hand-object system is NOT affected by the grasped object weight.

(1 mark)

- True
- False

viii) A non-symmetric post-contact reconfiguration of the hand-object system is a type of extrinsic manipulation.

(1 mark)

- True
- False

ix) A post-contact, parasitic reconfiguration of the hand-object system can only be symmetric.

(1 mark)

- True
- False

x) Large angles of the tendon rerouting on metal pins - in the tendon routing system - lead to decreased friction in the actuation of the finger.

(1 mark)

- True
- False

xi) The post-contact reconfiguration of the hand-object system on the table surface prior to the grasp depends on the table material.

(1 mark)

- True
- False

xii) In a tendon-driven hand that is actuated by a single servo motor that has a pulley attached onto the motor horn, the larger the pulley of the motor, the larger the pulling force exerted on the tendon.

(1 mark)

- True
- False

3.2 In the context of tendon-driven, compliant prosthetic grippers and hands that are controlled using muscle-machine interfaces:

i) Briefly explain what method leads to the easiest to develop and most affordable muscle machine interfaces, Electromyography, Forcemyography, or Sonomyography? (2 marks)

ii) Describe Forcemyography and Sonomyography. (2 marks)

iii) In Forcemyography-based control of a human-like prosthesis, explain how you could use regression and classification methods, giving an example of each. (3 marks)

iv) Briefly describe a grasp quality measure that depends on:
   a) the location of the object centre of mass. (1 mark)
   b) the location of the contact points of the gripper or hand. (1 mark)

v) Briefly explain what bioinspiration is and give an example of a bio-inspired mechanism of a prosthetic finger. (3 marks)

vi) What is the Electromechanical Delay in the context of human muscles and what does this delay depend on? (2 marks)

3.3. In the context of Machine Learning.

i) Explain what random forests are, for what types of supervised learning can be used, and their method of voting. (2 marks)

ii) What is the difference between supervised and unsupervised learning. (2 marks)